GRADE 5 · MATHEMATICS

INDICES & POWERS

Fun with Numbers and Exponents!

🗐 In this lesson you will learn:

- What indices (powers) mean
- Key vocabulary: base, exponent, index
- The 5 Laws of Indices
- Worked examples for each law
- 20 practice MCQ questions with an answer key

What Are Indices & Powers?

Indices (or powers) tell us how many times to multiply a number by itself. It's like a shortcut for repeated multiplication!

Example 1: 3^2 means $3 \times 3 = 9$. The small number 2 is called the **exponent** or **index**.

Example 2: 2^3 means $2 \times 2 \times 2 = 8$.

💡 Think of it like this: when you see a big number with a small superscript number above it, that small number tells you how many times to multiply the big number by itself! This makes math easier and helps us write big numbers in a simple way.

Vocabulary & Notation

Base
The number being multiplied (e.g., 3 in 3^2)

Exponent
How many times we multiply (e.g., 2 in 3^2)

Example
$3^2 = 3 \times 3 = 9$ (3 is the base, 2 is the exponent)

Also called
Index or power – all mean the same thing!

Understanding Powers

Powers tell us how many times to multiply a number by itself. Let's explore some fun examples with numbers 2 and 3.

$2^1 = 2$ (just the number itself)

$3^1 = 3$ (any number to power 1 equals itself)

$2^2 = 2 \times 2 = 4$

$2^3 = 2 \times 2 \times 2 = 8$ and $3^2 = 3 \times 3 = 9$

Law of Indices #1 – Product Law

When multiplying powers with the same base, add the exponents! This makes multiplication much easier.

Rule: $a^m \times a^n = a^{(m+n)}$

Example: $2^2 \times 2^3 = 2^{(2+3)} = 2^5 = 32$

Try: $3^3 \times 3^2 = 3^{(3+2)} = 3^5 = 243$

Law of Indices #2 – Quotient Law

The Quotient Law helps us when dividing powers with the same base. Instead of doing long division, we simply subtract the exponents!

Rule: $a^m \div a^n = a^{(m-n)}$, where m is greater than n

Example: $5^4 \div 5^2 = 5^{(4-2)} = 5^2 = 25$

⚠ Remember: The base must be the same, and m must be greater than n!

Law of Indices #3 – Power of a Power

When a power is raised to another power, multiply the exponents!

Rule: $(a^m)^n = a^{(m\times n)}$

Example: $(3^2)^3 = 3^{(2\times 3)} = 3^6 = 729$

💡 Think of it like doing the multiplication m times, n times each!

Law of Indices #4 – Zero Exponent Law

Any number raised to the power of zero equals 1 (where the number is not zero).

Rule: $a^0 = 1$ (where $a \neq 0$)

$7^0 = 1$ $10^0 = 1$ $5^0 = 1$

⚠ This rule only works if your base number is not zero! 0^0 is undefined.

Law of Indices #5 – One Exponent Law

Any number to the power of 1 equals the number itself.

Rule: $a^1 = a$

$5^1 = 5$ $100^1 = 100$ $7^1 = 7$

Dividing Powers – More Examples

Example 1: $8^5 \div 8^3 = 8^{(5-3)} = 8^2 = 64$

Example 2: $10^4 \div 10^1 = 10^{(4-1)} = 10^3 = 1{,}000$

Practice Time! 20 MCQ Questions

Circle the correct answer for each question. Good luck! 🌟

1. What is 4^2?

A) 8
B) 16
C) 12
D) 4

2. In 5^3, what is the base?

A) 3
B) 5
C) 8
D) 15

3. In 7^4, what is the exponent?

A) 7
B) 11
C) 4
D) 28

4. What does 2^3 mean?

A) $2 + 2 + 2$
B) 2×3
C) $2 \times 2 \times 2$
D) 3×3

5. What is 6^1?

A) 0
B) 1
C) 6
D) 36

6. What is 9^0?

A) 0
B) 1
C) 9
D) 90

7. What is 3^3?

A) 9
B) 27
C) 6
D) 33

8. Using the Product Law, what is $2^2 \times 2^3$?

A) $2^5 = 32$
B) $2^6 = 64$
C) 4^5
D) $2^1 = 2$

9. Using the Product Law, what is $3^3 \times 3^2$?

A) $3^6 = 729$
B) $3^5 = 243$
C) $3^1 = 3$
D) 9^5

10. Using the Quotient Law, what is $5^4 \div 5^2$?

A) $5^2 = 25$
B) $5^6 = 15{,}625$
C) $5^1 = 5$
D) 1^2

11. Using the Quotient Law, what is $8^5 \div 8^3$?

A) $8^2 = 64$
B) 8^8

C) $8^1 = 8$
D) 8^{15}

12. What is $(3^2)^3$ using the Power of a Power law?

A) $3^5 = 243$
B) $3^6 = 729$
C) 3^9
D) 9^3 only

13. What is $10^4 \div 10^1$?

A) $10^3 = 1,000$
B) $10^4 = 10,000$
C) 10^5
D) 1

14. Which rule is used for $a^m \times a^n$?

A) Subtract exponents
B) Add exponents
C) Multiply exponents
D) Divide exponents

15. Which rule is used for $(a^m)^n$?

A) Add exponents
B) Subtract exponents
C) Multiply exponents
D) Keep exponents the same

16. What is 0^0?

A) 0
B) 1
C) Undefined
D) 10

17. What is 100^1?

A) 1
B) 10
C) 100
D) 1,000

18. What is 2^4?

A) 8
B) 16
C) 6
D) 12

19. For the Quotient Law $a^m \div a^n$, which condition must be true?

A) m must equal n
B) n must be greater than m
C) m must be greater than n
D) a must be zero

20. What is $5^2 \times 5^0$?

A) 25
B) 0
C) 1
D) 125

✅ Answer Key

1. B
2. B
3. C
4. C
5. C
6. B
7. B
8. A
9. B
10. A
11. A
12. B
13. A

14. B
15. C
16. C
17. C
18. B
19. C
20. A

Great job learning about Indices & Powers! Keep practicing! 🌟